Know the network.

A slide-first guide to the material supplied through October 20, 2026.

MIDTERM STUDY GUIDE

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CONCEPTS + CALCULATIONS + DESIGN REASONING

Use the original sample for question style, the slides for course emphasis, and the worked examples here to practise explaining your reasoning. Practice questions are original study material, not an official answer key.

Study unit	Textbook mapping	Guide pages
1. Foundations and design goals	Chapter 1	2-3
2. Application architectures	Chapter 2	4-5
3. Physical layer and media	Chapter 3	6-7
4. Data link and efficiency	Chapter 4	8-9
5. Network and transport	Chapter 5	10-13
6. Wired and wireless LANs	Chapter 7	14-18

Numbering matters
These are six study units, **not textbook Chapters 1-6**. The LAN title slide explicitly says Chapter 7 and that Chapter 6 is covered later. No Chapter 6 lecture was supplied.

Study order: understand the distinctions, work the calculations without looking, explain the scenario decisions, then attempt Set C with feedback held until the end. Finish with the recall check on pages 19-20.

Sources: uploaded Lectures 1-5; Assignment 1 themes; earlier sample midterm.

Design goals and structure

Name the goal, identify the mechanism, then state the practical limit.

Goal	What it asks	Evidence of a useful change
Performance	How quickly and how much work can the network handle?	Lower response time; higher useful throughput; less bottleneck load.
Availability	Is the required service operational when users need it?	A working backup/failover path can keep service running after failure.
Scalability	Can it accommodate growth without unacceptable deterioration?	Capacity headroom, modular expansion, or more appropriately deployed servers.

Security, manageability, usability and affordability also matter, but they do not replace the three key goals emphasized in this course. A faster link does not by itself protect against a failed switch. More servers do not guarantee availability unless they can actually take over the relevant service.

Structured design: three different jobs



The core condition here follows the lecture: building backbones across two or more adjacent buildings.

Access: connects clients, servers and printers into LANs, mainly using switches and possibly APs. **Distribution:** interconnects those LANs in the building. **Core:** interconnects building backbones in a campus-style network. Do not add a core merely because one floor has several departments.

> **Do not mix two questions**
> The organizational benefit of structured design is a clear, modular hierarchy that makes design, expansion and management easier. To claim better performance, availability or scalability, explain the **specific implementation** that produces that benefit.

Geographic scope: a LAN covers a defined local area; a backbone connects networks; a WAN spans larger distances and often relies on leased/carrier circuits. **Access:** an intranet serves authorized insiders; an extranet gives selected invited outsiders controlled access.

Layers, addresses and overhead

Protocol layers and structured-design layers describe different things.

Internet layer	Main job	Recognize
Application	Network services used by programs	HTTP, SMTP, DNS; user/application data
Transport	End-to-end application communication; segmentation/reassembly	TCP segments; UDP datagrams; port numbers
Network	Logical addressing and routing	IP addresses; IP packets/datagrams
Data link	Local device-to-device delivery; framing and error checks	Ethernet frames; MAC addresses
Physical	Represent and transmit bits on a medium	Signals, connectors, cable/radio specifications

Vertical communication is between adjacent layers in one system. **Horizontal communication** is the logical relationship between corresponding layers on different systems. A protocol is a set of communication rules; some rules are implemented in hardware, others in software.

Encapsulation: each layer adds its own information



The receiving system removes and interprets the relevant control information as data moves upward.

Layering makes components easier to replace and troubleshoot, provided interfaces remain compatible. Its cost includes header/trailer bytes and encapsulation processing. It does not automatically make communication faster.

> **Two frequent multiple-choice traps**
> **IP is network layer, not transport.** Also, the word **datagram** alone is ambiguous: it can describe UDP at transport or IP at network. Identify the protocol. The sample question does not name it, so its study key is not an official resolution.

In the simplified forwarding model, routers process physical, data-link and network information. Application and transport processing for the end-to-end exchange happens at the hosts. Routers can still run their own management applications.

Place the application work

Application architecture allocates functions; network architecture arranges network components.

TPS function	What it does
Data storage	Stores and retrieves application data.
Data access logic	Forms/performs queries to access a required subset of data.
Application logic	Performs the work or business rules of the application.
Presentation logic	Handles the user interface.
Services logic	Handles automated access from one application to another, if needed.

A transaction is a business event such as an order, payment or enrolment. A TPS captures and processes the event, updates database records and produces output. The lecture distinguishes this from media delivery and communication applications such as online radio or videoconferencing.

Architecture	Allocation in the lecture	Central implication
Host-based	Host performs all five functions; terminals exchange interaction messages.	Central processing; costly upgrades in large increments; possible bottleneck.
Client-based	Server stores data. Client performs access, application and presentation logic; no services logic in the model.	Queries require moving data to the client; excessive circuit traffic and limited server capacity.
Client-server	Client presents the interface. Server handles storage/access/services. Application logic may be on either side or split.	Server-side filtering avoids unnecessary data movement and permits work distribution.

Explain why, rather than memorizing a label
Client-based designs perform the query at the client, so data must cross the network for processing. Client-server can query at the server and return the needed subset. That is the mechanism behind the performance argument.

Tiers, clients and cloud

Select architecture using the three goals, then discuss its trade-offs.

Arrangement	Where the work goes
Two-tier	Client and server roles; application logic can be at the client, server or split. The slide gives fewer than 50 users as a rule of thumb, not a universal limit.
Three-tier	Client: presentation. Application server: application/services logic. Database server: storage/data access.
n-tier	More functional server roles, such as a web tier. Can distribute load and support growth, but adds communication and complexity.

Thin client: little or no application logic resides locally. Application maintenance can be concentrated on servers. **Thick client:** most or all application logic runs locally; suitable when users need their own micro-computing applications. Thin/thick describes work placement, not physical computer size.

Middleware sits between applications, translates across software systems and manages messages. It can hide some server/network changes from clients; the lecture associates it with services logic.

Cloud model	Provider supplies (lecture model)	Customer retains
SaaS	Application and supporting hardware/software.	Use/configuration of the service.
PaaS	Programming tools, OS/DBMS platform and hardware.	Developing and owning the application.
IaaS	Infrastructure hardware: compute, storage and networking.	Application and other software responsibilities.

A defensible architecture answer
Name the architecture; state which functions run where; explain performance and growth mechanisms; identify the failure/backup arrangements needed for availability. Architecture alone does not guarantee zero downtime or unlimited capacity.

The lecture permits host-based designs for some large-volume main-line TPS, generally recommends client-server otherwise, and treats cloud as an outsourcing variation. Its P2P file-sharing example has peers acting as both client and server, with a central service helping locate files.

Circuits and media

Keep physical medium, logical sharing and data-flow direction separate.

Data flow	Meaning	Lecture example
Simplex	One direction only.	TV/GPS reception.
Half-duplex	Both directions, one at a time.	Shared Wi-Fi communication.
Full-duplex	Both directions simultaneously.	Dedicated switched Ethernet links.

Point-to-point: a dedicated logical connection between two endpoints, as with a switched link. Separate links can carry traffic concurrently. **Multipoint:** several devices share the medium, as with a hub or AP radio. Shared access requires coordination.

Medium	What to remember	Typical use / limitation
Twisted pair	Insulated copper pairs; twists reduce interference. STP adds shielding. Relatively inexpensive.	Typical desktop/access channels up to 100 m without a repeater; attenuation matters.
Coaxial	Central conductor with surrounding shield; shielding reduces interference.	Cable TV/Internet; not the common organizational LAN choice in these slides.
Fiber	Light through glass/plastic core; resistant to electromagnetic interference.	Backbones/longer spans; higher cost and compatible optics required.
Radio	Wireless electromagnetic transmission; clients need radio NICs.	Mobility and difficult-to-cable locations; shared airtime, interference and security considerations.

Cable question traps
Coax reduces interference through **shielding**. Twisted pair uses **twisting**, with additional shielding in STP. A conventional four-pair Ethernet cable contains **eight individual conductors**; pairs and conductors are not the same count.

Choose media and conversion

A good recommendation names the constraint it solves.

Constraint	Reasonable direction	Reasoning
Short fixed desktop runs	Twisted pair, within its supported channel distance.	Lower cost; common access-layer medium.
Long span / electrical interference	Appropriate fiber grade and optical interfaces.	Longer reach and resistance to electromagnetic interference.
Moving users / difficult cabling	WLAN overlay with a site survey.	Mobility or avoided cabling cost; capacity/security still need design.
Microwave path	Line-of-sight planning with suitable antennas.	Obstructions, weather and Earth curvature can limit the link.

The slides compare multimode (~500 m), graded-index multimode (~1 km) and single-mode systems (~100 km or more). These are **not universal ratings for every optic**: actual reach depends on fiber, standard and transceivers.

Do not confuse the two conversions

Situation	Component	What changes
Digital computer data over an analog circuit	Modem at each end.	Digital data is modulated for analog transmission and recovered at the receiver.
Analog voice over a digital network	Codec at each end, often inside a VoIP phone.	Voice is digitized for transport and converted back for listening.
Digital data over a digital network	Compatible NICs, signals and standards.	Devices agree on how digital information is represented and transmitted.

> **One distinction at a time**
> Serial transmission sends bits sequentially on a serial channel; parallel transmission uses multiple channels together. That is different from **simplex/duplex**, which describes direction, and **point-to-point/multipoint**, which describes sharing.

Before recommending equipment, check the supported rate, channel length, connector/media compatibility, cost, interference, security and mobility needs. A nominally faster component cannot overcome an incompatible link.

Control access and detect errors

Data link handles framing, local delivery, medium access and error checks.

Access approach	Operation	Trade-off
Contention	Stations try when the shared medium is available, without central permission.	Less coordination overhead at low load; collisions/retries can become costly at high load.
Controlled access	Permission or turns are coordinated, e.g. polling or request-based control.	Coordination adds overhead but can reduce the cost of contention under heavy traffic.

CSMA/CD: Carrier Sense means listen before transmitting. Multiple Access means multiple devices can access the medium. Collision Detection means detect a collision while transmitting, stop and retry after a waiting interval. CD does not repair lost characters and is not the same as waiting before transmission.

A hub produces a shared half-duplex collision domain. A dedicated full-duplex switched link does not require collision arbitration on that link. Do not apply shared-hub collision behaviour to all modern switched Ethernet.

Error control is broader than error detection

The lecture defines error control as prevention + detection + correction. Causes include impulse noise, crosstalk, echo and attenuation. Prevention choices include shielding, suitable cable placement, good connections and repeaters where appropriate.

> **CRC-32: what happens to a bad frame?**
> The sender includes a check value in the frame check sequence (FCS). The receiver checks for corruption. **CRC detects; it does not reconstruct damaged bits.** Ethernet discards bad frames. TCP can recover by retransmission; UDP itself does not promise recovery.

A passing CRC is strong evidence of no detected transmission corruption, not a mathematical guarantee that every possible error is impossible. The FCS, source/destination MAC addresses and other link-control fields are overhead relative to application payload.

Efficiency: show the denominator

Use the overhead specified by the question; byte conventions can differ.

Core formula
Efficiency = useful payload / (useful payload + total overhead) x 100%
Add all stated transport, network and link overhead once per transmission. Keep numerator and denominator in the same units.

Worked example: overhead at three layers

An update contains **300 user bytes**, a 20-byte TCP header, a 20-byte IPv4 header and 40 bytes of link overhead. Assume one update per transmission and include only the stated overhead.

Step	Calculation
Total overhead	20 + 20 + 40 = 80 bytes
Total transmitted	300 + 80 = 380 bytes
Useful percentage	300 / 380 x 100 = **78.95%**
At 4 transmissions per second	1,200 useful bytes / 1,520 total bytes per second = **78.95%**
At 8 transmissions per second	2,400 useful / 3,040 total = **78.95%**

Frequency: doubling the number of identical messages doubles useful data and overhead per second, leaving the percentage unchanged. The amount of circuit capacity consumed still increases.

Smaller payload: if useful data falls to 30 bytes while overhead stays 80, efficiency becomes 30 / 110 = **27.27%**. The fixed overhead consumes more of the transmission.

Batching: two 300-byte updates in one transmission with one 80-byte overhead cost give 600 / 680 = **88.24%**. That can improve efficiency but waiting to batch data may increase delay.

Three exam traps
Do not divide payload by overhead alone. Do not subtract headers from payload that was already described as user data. Do not substitute a memorized Ethernet overhead: the lecture and assignment use different accounting examples.

Nominal bit rate, useful-data efficiency and application response time are related but different. A more efficient large frame does not guarantee the best latency or automatically override a standard frame-size limit.

Transport and address services

Identify the protocol before naming its layer or job.

TCP	UDP
Connection-oriented; uses connection setup.	Connectionless; no TCP-style connection handshake.
Reliable, ordered byte-delivery mechanisms; retransmission and flow control.	No delivery/order/retransmission guarantee supplied by UDP itself.
Transport PDU: segment; header is at least 20 bytes.	Transport PDU: datagram; standard header is 8 bytes.

TCP/UDP headers contain source and destination **port numbers** to identify endpoint applications/processes. They are logical identifiers, not physical Ethernet sockets. Application design determines protocol use; administrators configure network QoS policies with compatible equipment/software.

Service	What it does	Avoid this confusion
DHCP	Supplies leased IP configuration, reducing manual setup and allowing reuse.	Dynamic addressing is not dynamic routing.
DNS	Resolves a readable hostname to an IP address.	It does not resolve the local next-hop MAC address.
ARP	Finds a local IPv4 next hop's MAC address.	For a remote subnet, resolve the gateway's MAC, not the remote host's MAC across the Internet.

Address sizes: IPv4 = 32 bits (four 8-bit octets); IPv6 = 128 bits; Ethernet MAC = normally 48 bits, shown in hexadecimal. IP provides logical location and routing; MAC supports local frame delivery.

> **Read slide shorthand carefully**
> Lecture 4 slide 9 lists UDP overhead as 4-8 bytes and TCP as 20-24. Standard UDP headers are 8 bytes; TCP headers start at 20 and can grow with options. In arithmetic, use the sizes explicitly supplied by the question.

Public/private describes routing scope. **Static/dynamic** describes assignment. These are separate properties: a public address can be dynamically assigned and a private address can be static.

Subnetting: the reliable method

Borrow bits for subnets; leave enough bits for usable hosts.

Original class	Default prefix	Default mask	Host bits
A	/8	255.0.0.0	24
B	/16	255.255.0.0	16
C	/24	255.255.255.0	8

Four quantities to calculate
Borrowed subnet bits: s = new prefix - original prefix
Remaining host bits: h = 32 - new prefix
Subnets: 2^s
Usable hosts per subnet: 2^h - 2

The host subtraction excludes the all-zero network address and all-one broadcast address. The lecture uses **2^s for subnets**, without subtracting two. It keeps at least two host bits; use that conventional model for these exercises.

Mask octet	Binary	Leading 1-bits
0	00000000	0
128	10000000	1
192	11000000	2
224	11100000	3
240	11110000	4
248	11111000	5
252	11111100	6
254	11111110	7
255	11111111	8

A valid conventional mask has consecutive 1-bits on the left followed by 0-bits. **255.255.224.0 = /19**. Masks such as 255.255.32.0 or 0.0.0.255 fail this contiguous-mask rule.

Subnetting: work both directions

First identify whether the question asks for subnets, hosts, borrowed bits or the mask.

A. Required subnet count -> mask and hosts

A Class B /16 block needs at least **90 equal-size subnets**. Six bits give 64, insufficient; seven give 128, sufficient. Borrow **7**. New prefix = /23; mask = **255.255.254.0**. Nine host bits remain, giving **510 usable hosts** per subnet. After allocating 90 subnets, 38 are spare.

B. Required hosts -> mask and subnet count

A Class C /24 block needs at least **45 usable hosts per subnet** while maximizing subnet count. Five host bits give 30, insufficient; six give 62. Keep **6 host bits**. New prefix = /26; mask = **255.255.255.192**. Borrow two bits, giving **4 subnets**.

C. Given mask -> distinguish host and subnet bits

A Class B block with **255.255.255.248** uses /29. Borrowed bits = 29 - 16 = **13**, giving **8,192 subnets**. Host bits = 32 - 29 = 3, giving **6 usable hosts** each. A Class C /24 with the same /29 mask borrows only five bits: **32 subnets**, still six usable hosts each.

What changes?	What stays / changes?
More borrowed bits inside a fixed block	More subnets; fewer usable hosts per subnet.
Same final mask, different original block	Same host capacity per subnet; different number of subnets within the original allocation.
More hosts than a subnet can hold	Requires another subnet or a revised allocation; spare subnet count does not enlarge one existing subnet.

The class label is not the final host count
A Class A address with a /28 mask has only 4 host bits left: **14 usable hosts**. Do not use the original 24 Class A host bits after a supplied mask has changed the boundary.

Self-check: Class C with /28 -> four borrowed bits, 16 subnets, 14 usable hosts. Class B with /26 -> ten borrowed bits, 1,024 subnets, 62 usable hosts.

Routing and the next hop

An IP destination and a local frame destination do not always name the same device.



For a remote subnet, the first Ethernet frame addresses the local gateway, while the IP packet identifies the remote destination.

A router consults a routing table to choose a next hop or outgoing interface. Routing moves packets between subnets. A Layer 2 switch forwards local frames using MAC information; it does not decide an inter-subnet path simply because the frame contains an IP packet.

Route type	How it is maintained	What to remember
Static	Administrator configures routes.	Useful for simpler/predictable networks; does not learn changes through route exchanges.
Dynamic	Routing protocols exchange information and update routes.	The selected path depends on protocol metrics, not simply physical cable length.

The lecture contrasts distance-vector and link-state approaches. Its RIP example uses hop count; OSPF uses cost. Do not generalize that every distance-vector protocol uses only hop count. The detailed routing-protocol comparison on slide 26 explicitly says it is not covered by exams.

Private IPv4 range	Entire reserved range
10/8	10.0.0.0 - 10.255.255.255
172.16/12	172.16.0.0 - 172.31.255.255
192.168/16	192.168.0.0 - 192.168.255.255

NAT in the lecture model
Private internal IPv4 addresses are not globally routed directly on the public Internet. NAT at the boundary translates them for Internet access. It is separate from DHCP leases, ARP resolution and DNS name lookup.

Hubs, switches and topologies

Physical topology describes the layout; logical topology describes the data flow.

Feature	Hub	Switch
Physical layout	Often star-shaped around the hub.	Often star-shaped around the switch.
Logical operation	Shared bus; repeats signals to other ports.	Logical star with dedicated point-to-point links.
Layer in lecture model	Layer 1.	Layer 2; some products also have Layer 3 functions.
Duplex / collisions	Shared half-duplex collision domain.	Dedicated full-duplex links remove collision arbitration on each link.
Forwarding basis	Does not use a MAC table to select one output.	Looks up destination MAC information in its forwarding table.

The switch description assumes a learned destination in its forwarding table. Do not turn the simplified idea into a claim that switches never send a frame to multiple ports: broadcasts and some other traffic are exceptions. The main exam distinction is shared repeating versus destination-based switching.

Switching mode	When forwarding begins	Trade-off
Cut-through	After destination information is read.	Low forwarding delay, before a full-frame error check.
Store-and-forward	After the full frame arrives and is checked.	Can reject a bad frame before forwarding; additional receive/check delay.
Fragment-free	After the first 64 bytes arrive.	A compromise; not a full-frame FCS check.

Do not confuse switching mode with duplex
Cut-through/store-and-forward describe when a switch begins forwarding. Full-duplex describes simultaneous transmission directions on a link. Neither term describes how many host bits are in the subnet mask.

A NIC supplies a network interface and MAC identity for wired or wireless communication. An AP connects wireless clients into the wired LAN. An AP is a different function from a router, even when a consumer product combines both.

Ethernet rates and bottlenecks

A nominal rate is a capability, not a promise of useful application throughput.

Label / family	Interpretation	Design implication
10BASE-T	10 Mb/s; baseband; twisted-pair medium.	An older Ethernet example in the slides.
100BASE-TX	Fast Ethernet, 100 Mb/s nominal rate.	Must match supported NIC/switch rates and media.
1000BASE-T	Gigabit Ethernet, 1,000 Mb/s = 1 Gb/s.	Often used for switched access; still needs compatible cabling.
Fiber Ethernet variants	Optical interfaces with different supported rates/reaches.	Use the actual standard/optic rating, not a generic fiber maximum.
Multi-speed interfaces	Support compatible negotiated/configured link rates.	A 100 Mb/s endpoint does not become 1 Gb/s merely because another switch port is faster.

Bits versus bytes: 1 Gb/s is 1,000 Mb/s nominally, not 1,000 MB/s. Converting bits to bytes divides by eight before overhead. Avoid mixing units in a throughput explanation.

> **Worked bottleneck scenario**
> Six clients can each connect at 1 Gb/s but share one 1 Gb/s uplink. They do not obtain six independent 1 Gb/s paths through that uplink. Concurrent useful traffic is constrained by its capacity, protocol overhead and traffic mix. Faster client ports alone do not remove that shared bottleneck.

Connect the fix to the measured resource

If server CPU/storage is limiting, consider server resources or additional appropriately deployed servers. If the uplink is saturated, increase suitable circuit capacity or reduce the traffic crossing it. If radio airtime is busy, investigate WLAN channels, placement, load and backhaul.

Goal links: more supported capacity may improve performance and leave growth headroom. Availability needs failure handling, not just a higher speed label. The lecture also recommends securing switches in locked closets; that is a security measure, not a throughput calculation.

Wireless operation and access

802.11 integrates with Ethernet; it is not the same standard.

A WLAN provides mobility and access where cabling is difficult or expensive. The lecture recommends wireless as an **overlay for mobile users** and avoids it for servers. A client must select/operate on an AP's channel to exchange data. Clients and AP radios share airtime.

Concept	Meaning
Physical / logical topology	The lecture describes a physical star and logical shared bus.
CSMA/CA	Carrier-sensing with collision-avoidance procedures, including random backoff.
Acknowledgements and retries	For ordinary acknowledged unicast exchanges, an absent ACK can trigger retry; the same rule is not applied to every broadcast/multicast frame.
Hidden nodes	Two clients can reach the AP but cannot adequately sense one another.
RTS/CTS	A request/clear-to-send exchange can help coordinate access and reduce hidden-node effects.



Hidden-node case: the clients may be unable to sense one another even though each can communicate with the AP.

A slide simplification to recognize
Slide 13 groups RTS/CTS under PCF. In standard terminology, DCF is contention-based and can use RTS/CTS; PCF is AP-controlled polling. Learn the lecture's intended contrast (distributed contention versus coordinated access), but do not memorize "RTS/CTS exists only in PCF."

Collision avoidance is not a guarantee of zero collisions. An AP does not transform shared wireless airtime into a dedicated full-duplex wired link. Wi-Fi acknowledgements also do not make UDP into TCP.

Plan wireless capacity and security

Coverage, channel capacity and secure access are separate design questions.

Standard family in slides	Band association to recognize
802.11b/g	2.4 GHz.
802.11a/ac	5 GHz.
802.11n	2.4 and/or 5 GHz, depending on the implementation.
802.11ax	Wi-Fi 6 family; 6 GHz operation is associated with Wi-Fi 6E equipment.

These are recognition points from the course; avoid treating the slide's "most common" or "most advanced" labels as permanent facts. Raw rates depend on channel width, streams and implementation. Actual useful rates also depend on user load, distance, obstacles, interference and overhead.

Channel planning: suitably planned non-overlapping channels allow more concurrent use while limiting interference. Adding same-channel APs close together does not automatically add independent capacity. For simultaneous 2.4 GHz 802.11n and 5 GHz 802.11ac support, select AP radios capable of **concurrent dual-band operation**.

Control	Purpose / limitation
Encryption	Protects transmitted data. Recognize the WEP -> WPA -> WPA2 -> WPA3 progression; WEP is weak. WPA2 uses AES in the lecture comparison.
Authentication	Controls access using identity, credentials and permissions.
MAC filtering	Checks permitted device addresses, but addresses can be spoofed. It is not a substitute for strong authentication/encryption.

Site-survey checklist
Coverage and floor plan; users/capacity needs; interference; AP count and locations; non-overlapping channel assignments; switch/backhaul locations; power/PoE; controller needs; guest traffic requirements. Coverage alone does not prove adequate capacity.

PoE can supply AP power through suitable switching/cabling. An AP controller can centralize management of many APs; it does not remove all radio or backhaul limits.

Make a defensible recommendation

Use the sequence: observation -> bottleneck -> intervention -> goal -> limitation.

Observed constraint	Supply-side response	Demand-side response
Server CPU/memory/storage	Tune NOS settings; improve relevant CPU/RAM/storage; distribute work with suitable extra servers.	Reduce unnecessary processing and shift suitable work to capable clients.
Circuit/uplink capacity	Upgrade compatible links; use switched access; segment LANs appropriately.	Schedule large transfers/backups off-hours; block unnecessary traffic.
Wireless contention/coverage	Survey AP placement, channels, radios and backhaul; use suitable compatible equipment.	Move suitable fixed users to wired access; manage traffic and peak demand.

The lecture uses high server CPU utilization (around 80% or more) as a bottleneck warning. Treat it as a signal to investigate alongside workload and other measurements, not proof that every busy CPU must be replaced.

A model explanation

> **Scenario: quiet server, saturated uplink**
> "The shared uplink is the observed bottleneck. Increasing compatible uplink capacity or moving backups off-hours can reduce congestion and improve response time. The first adds supply; the second reduces peak demand. Extra link capacity may leave growth headroom, but neither action alone provides a backup path if the switch fails."

Questions to ask before choosing hardware

Where is the restriction? CPU, storage, access link, uplink, radio or backhaul?
What is required? Number of simultaneous users, traffic volume, mobility and coverage?
What must interoperate? NICs, media, connectors, port rates, wireless bands and power?
Which goal changes directly? Explain the mechanism rather than claiming all three improve automatically.

The backup LAN slide introduces data-center server clusters, load balancing, virtualization, SAN versus NAS, and an e-commerce edge. Recognize them if covered in class: a SAN is a dedicated storage network; NAS is a storage server attached to an ordinary LAN. The SOHO section is explicitly not covered in this lecture.

Can you explain it unaided?

Write short answers first; compare your reasoning with the checkpoints on the next page.

1. Three technical goals

A faster uplink is installed but the business still has one access switch. Explain which goal can improve directly and which failure problem remains.

2. Application placement

Why can client-based database queries overload circuits? Explain how moving data access logic to a server changes the traffic.

3. Media selection

Compare twisted pair, fiber and radio for fixed desktops, a long electrically noisy route and mobile staff.

4. Errors and recovery

A corrupted frame is discarded. Explain the separate jobs of CRC, Ethernet and a reliable transport protocol.

5. Subnet calculation

A Class B /16 block requires 90 equal-size subnets. Derive borrowed bits, mask, usable hosts and remaining subnet capacity.

6. Efficiency calculation

A transmission carries 300 useful bytes and 80 overhead bytes. Compare its efficiency with a 30-byte payload and the same overhead.

7. Switching

Compare a hub and switch physically and logically. Then compare cut-through and store-and-forward switching.

8. WLAN diagnosis

Signal coverage seems adequate, but users slow down when a room fills. Explain why adding another same-channel AP is not automatically the right fix.

Next: attempt Set C with feedback held until the end. Review missed explanations and their slide references.

Checkpoints and last-pass traps

Use this page to identify what needs another active-recall attempt.

1. More uplink capacity can help performance and growth headroom; a failed single switch still interrupts service.

2. Client-side querying requires data movement; server-side filtering can return only needed results.

3. TP: inexpensive short access runs. Fiber: longer/noisy route with correct optics. Radio: mobility with channel/capacity/security planning.

4. CRC detects rather than repairs. Ethernet discards bad frames; TCP can recover by retransmission. UDP itself does not guarantee recovery.

5. Borrow 7 -> /23 -> 255.255.254.0 -> 128 subnets -> 510 usable hosts each -> 38 unallocated subnets after 90.

6. 300/380 = 78.95%; 30/110 = 27.27%. More fixed-overhead share means lower useful-data efficiency.

7. Similar physical star; hub shares/repeats, switch separates links. Cut-through starts earlier; store-and-forward checks the full frame first.

8. Shared airtime, interference and backhaul can limit capacity despite coverage. Survey channel/load constraints and match the intervention.

> **Last-pass memory checks**
> **Subnets:** 2^s. **Usable hosts:** 2^h - 2. **IP:** network layer. **TCP/UDP:** transport. **DNS:** name -> IP. **ARP:** local IPv4 next hop -> MAC. **DHCP:** leased settings. **CD:** collision detection. **CA:** collision avoidance. **Nominal rate:** not guaranteed user throughput.

Source map and scope

Lecture 1: foundations/goals/layers, slides 23-45. Application lecture: functions and architectures, slides 4-23. Physical/Data Link: media/access/errors/efficiency, slides 6-34. Network/Transport: protocols/addressing/subnetting/routing, slides 4-23. Wired/Wireless LAN: components, switching, wireless and performance, slides 3-20; backup recognition on slide 22. Chapter 6 is marked for later; Chapter 7 is included here.

Course limits: application textbook sections 2.3-2.6 and LAN SOHO section are marked not covered; the detailed network/routing table on slide 26 says it is not covered by exams. Instructor clarifications and material actually covered in class control scope. Original sample Q11, Q19 and Q23 have ambiguous wording; this guide does not invent an official key.

Technical clarifications: RFC 768 (UDP header), rfc-editor.org/rfc/rfc768; RFC 9293 (TCP), rfc-editor.org/rfc/rfc9293. Cisco wireless course, cisco.com/E-Learning/bulk/guest/celc/fwl/ch3/3_6_2/content.html (DCF/PCF and RTS/CTS). The guide prioritizes instructor slides; these references clarify protocol shorthand rather than add exam topics.